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B.Tech. Degree II Semester Regular Examination July 2024

EE/EC/CS/IT 23-200-0203B DIGITAL ELECTRONICS

(2023 Scheme)

Time: 3 Hours

Maximum Marks: 50

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Understand the fundamental boolean functions and basic building blocks of digital systems.

CO2: Design optimal digital circuits using basic building blocks.

CO3: Analyse basic digital circuits.

CO4: Understand HDL models of simple circuits.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

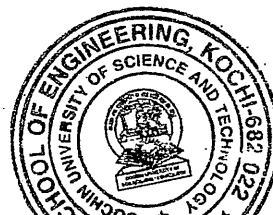
(Answer *ALL* questions)

	(5 × 2 = 10)	Marks	BL	CO	PO
I. (a) Simplify the following Boolean expression using Boolean algebra	2	L3	1	1,12	
(i) $AB + \overline{AC} + A\overline{B}C(AB + C)$					
(ii) $(X + \overline{Y} + XY)(X + \overline{Y})(\overline{X}Y)$					
(b) Find the minterm expansion of the following Boolean expression.	2	L3	1	1,12	
(i) $F(A, B, C, D) = \overline{A}(\overline{B} + D) + AC\overline{D}$					
(ii) $F(X, Y, Z) = X(\overline{X} + Y)(\overline{X} + Y + \overline{Z})$					
(c) Define the figure of merit of logic gates. The following are the voltage and current parameters of a 74 series logic gate, find fan out and figure of merit.	2	L3	2,3	1,12	
Propagation delay = 9 ns,					
Power dissipation per gate = 10 Mw,					
$I_{OH} = 800 \mu A$; $I_{IH} = 40 \mu A$; $I_{OL} = 48 \text{ mA}$; $I_{IL} = 1.6 \text{ mA}$.					
(d) Draw the timing diagram showing the loading of a 4-bit input data 0101 into a 4-bit SISO register with initial state of each flip flop in the register being logic 1.	2	L3	2,3	1,3,12	
(e) Draw the circuit of a 3-bit ring counter. Explain its working with timing diagram.	2	L2	2,3	1	

PART B

(4 × 10 = 40)

- II. (a) Determine the prime implicants of the Boolean function by using the tabulation method
 $F(a,b,c,d) = \sum m(1, 4, 6, 7, 8, 9, 10, 11, 15).$ 6 L3 1,2,3
- (b) Solve using a 3 variable K map 4 L3 1,2 1,2,3,12
- (i) $AB\overline{E}\overline{C} + ABC\overline{D} + ABCE + \overline{A}BC\overline{D} + \overline{A}BC$
- (ii) $m_1 + Dm_2 + m_3 + \overline{D}m_5 + m_7 + d(m_0 + Em_1 + \overline{E}m_6).$
- OR**
- III. (a) Design a logic circuit to produce a HIGH output if and only if the input, represented by a 4-bit binary number, is greater than 12 or less than 3. Develop the truth table and draw the logic diagram. 6 L6 1,2,3
- (b) Calculate by 2's complement method 4 L2 1,2,3
- (i) $(75.75)_{10} - (126.250)_{10}$
- (ii) $(A3.74)_{16} - (23.52)_8.$



(P.T.O.)

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		Marks	BL	CO	PO
IV.	Describe the working of a 4-bit look ahead carry adder. Clearly show the derivation of the equations and implementation of hardware.	10	L2	1,2,3	1
OR					
V. (a)	Implement the following function using 8×1 MUX $F = \sum m(0, 1, 2, 3, 4, 10, 11, 14, 15)$	5	L3	1,2,3	1,2,3
(b)	Realize JK flipflop using SR flipflop.	5	L2	1,2,3	
VI. (a)	Design and setup a MOD-6 ripple up-counter using positive triggered JK flipflop. Show the timing diagram of the counter showing the glitches in the diagram.	5	L6	1,2,3	1,2,3
(b)	Design and setup a circuit to generate the following sequence by indirect method.	5	L6	1,2,3	1,2,3
1 1 0 1 1 1 0					
OR					
VII.	Design and setup a synchronous sequential counter using JK flipflop and avoid lock-out condition for the sequence. $4 \rightarrow 6 \rightarrow 7 \rightarrow 3 \rightarrow 1 \rightarrow 4 \dots$	10	L6	1,2,3	1,2,3
VIII. (a)	Implement the following function using PLA $X(a, b, c, d) = \sum m(0, 2, 6, 7, 8, 9, 12, 13, 14)$ $Y(a, b, c, d) = \sum m(2, 3, 8, 9, 10, 12, 13)$ $Z(a, b, c, d) = \sum m(1, 3, 4, 6, 9, 12, 14)$	5	L3	1,2,3	1,2,3,12
(b)	Implement a full adder using PROM.	5	L3	1,2,3	1,2,3,12
OR					
IX. (a)	Draw and explain the working of a two input TTL NAND totem-pole configuration.	5	L2	1,2,3	1,12
(b)	Four 7-bit hamming codes are transmitted serially through a noisy channel. Detect and correct errors, if any, in the even parity hamming code.	5	L3	1,2	1,2,12

1001001011100111101100011011

Bloom's Taxonomy Levels

L2 - 23.33%, L3 - 42.22% L6 - 34.45%.
